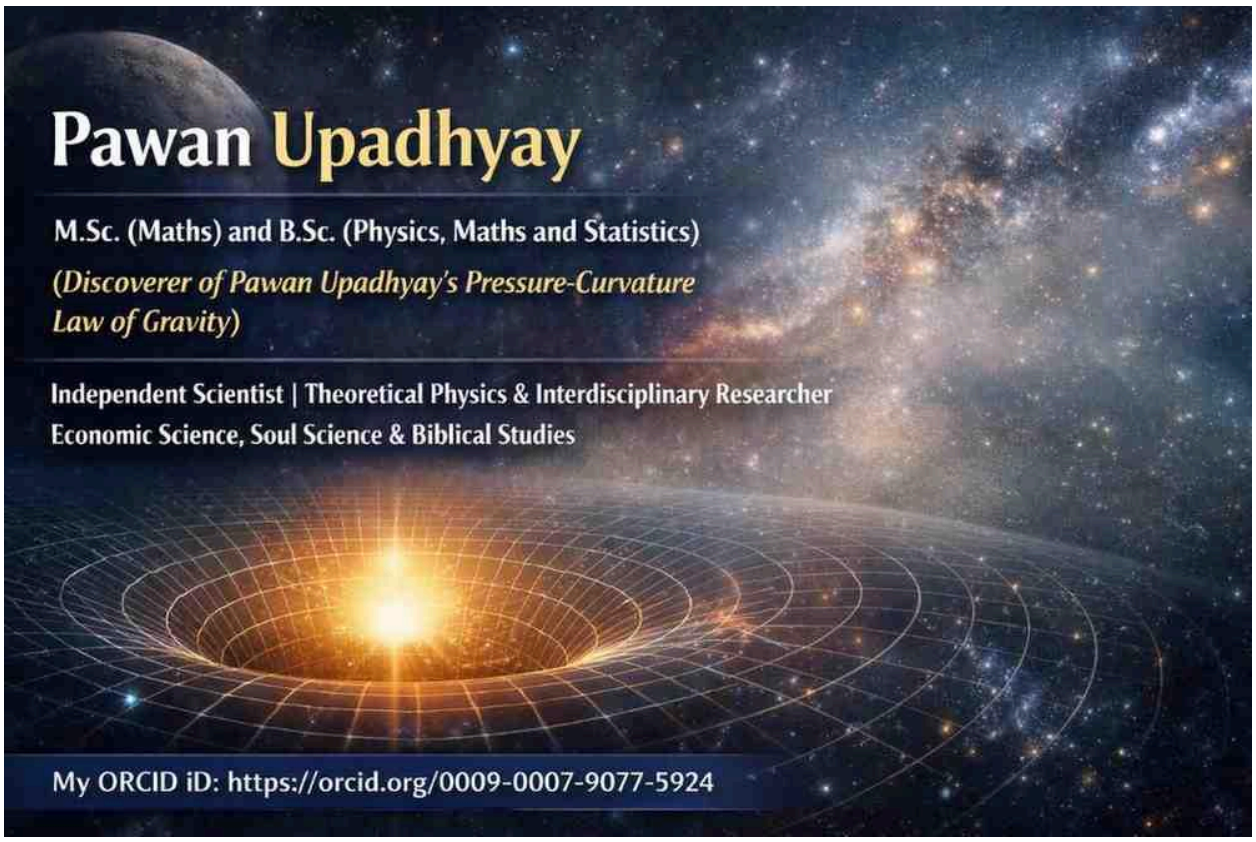
Note: Difference Between Scalar and Vector



1. Scalar Quantity

A scalar quantity is a physical quantity that has **magnitude only and no direction**. It is expressed by a numerical value and an appropriate unit, where applicable.

Examples of scalar quantities:

1. Mass — 5 kg
2. Time — 10 seconds
3. Temperature — 30°C
4. Distance — 20 metres
5. Speed — 60 km/h
6. Energy — 100 joules
7. Work — 50 joules
8. Pressure — 100 pascals

9. Volume — 2 litres
10. Density — 1,000 kg/m³

2. Vector Quantity

A vector quantity is a physical quantity that has **both magnitude and direction**. It can be represented by an arrow, whose length indicates magnitude and whose orientation indicates direction.

Examples of vector quantities:

1. Displacement — 5 metres east
2. Velocity — 60 km/h north
3. Acceleration — 9.8 m/s² downward
4. Force — 10 newtons upward
5. Momentum — 20 kg·m/s east
6. Weight — 50 newtons downward

3. Main Differences

Scalar Quantity	Vector Quantity
Has magnitude only.	Has both magnitude and direction.
Does not have a direction associated with the quantity.	Has a direction associated with the quantity.
Ordinary scalar addition applies to quantities of the same kind.	Vector addition follows vector rules.
Examples include mass, time, speed, energy, work and pressure.	Examples include displacement, velocity, acceleration, force and momentum.

4. Important Distinctions

Distance and displacement: Distance is a scalar, whereas displacement is a vector.

Speed and velocity: Speed is a scalar, whereas velocity is a vector.

Pressure and force: Pressure is a scalar quantity, whereas force is a vector quantity. Pressure acts on a surface and produces a force perpendicular to that surface in a fluid at rest, but pressure itself has no single direction.

5. Simple Everyday Example

Imagine you are walking from your home to a nearby shop.

- **Scalar example (Distance):** You walk 500 metres. We know how much distance you travelled, but no direction is specified.
- **Vector example (Displacement):** You walk 500 metres towards the east. We know both the magnitude (500 metres) and the direction (east).

Another example:

- **Scalar:** A car is moving at a speed of 40 km/h.
- **Vector:** A car is moving at a velocity of 40 km/h towards the north.

Remember: A scalar tells us *how much*, whereas a vector tells us *how much and in which direction*.

Pressure — Scalar Quantity (Everyday Example)

When you inflate a bicycle tyre, the air inside exerts pressure on the inner walls of the tyre. Suppose the tyre pressure is 200 kPa.

Pressure is a **scalar quantity** because it has magnitude but no single direction of its own. The force exerted by the air on the tyre walls acts perpendicular to each surface and is a vector quantity.

Remember: Pressure is scalar, but the force produced by pressure on a surface is a vector.

Conclusion

The fundamental difference between scalar and vector quantities is that a scalar has magnitude only, whereas a vector has both magnitude and direction. This distinction is essential for understanding physical quantities and applying the laws of physics correctly.